

CME295: Transformers and Large Language Models

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

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Course map

Source: SRC-01 (official syllabus page, inspected 2026-10-06). Dates and topic bullets are quoted from that page. Claim class per unit row: OFFICIAL-SYLLABUS = topic named on the syllabus, REQUESTED-BRANCH = requested extension not named on the syllabus.

#	Date	Lecture	Topics (syllabus)	Units	Claim
L1	Sep 25, 2026	Transformers	NLP background tasks, tokenization, embeddings, word2vec, RNN, LSTM, attention, transformer architecture, end-to-end example	U01	OFFICIAL-SYLLABUS
L2	Oct 2, 2026	Large Language Models	Transformer model families, LLM definition and architecture, mixture of experts, MHA, MQA, GQA, position embeddings (RoPE and variants), context length, temperature, sampling strategies	U02, U03	OFFICIAL-SYLLABUS
L3	Oct 9, 2026	LLM training	Pretraining, SFT, LoRA, preference tuning (RLHF, DPO), reasoning, on-policy distillation and variants, distillation to smaller models	U04, U05	OFFICIAL-SYLLABUS
L4	Oct 16, 2026	Reinforcement learning with ³	Mathematical conventions.	U05, U06	OFFICIAL-SYLLABUS

This build (first builder): U01-U05

- U01 anchors to L1 (all 12 concepts OFFICIAL-SYLLABUS).

- U02 anchors to L2: attention variants, MHA/MQA/GQA, RoPE and position

variants, encoder/decoder layouts, BERT derivatives are OFFICIAL-SYLLABUS. Attention approximation, cache differences beyond naming, complexity analysis, and failure cases are REQUESTED-BRANCH extensions (KV caching is named in L5, detailed cache math is the branch).

- U03 anchors to L2: LLM definition, MoE, context length, temperature,

sampling strategies are OFFICIAL-SYLLABUS. Prompting, ICL, demonstration selection, reasoning prompting, self-consistency, and evaluation limits are REQUESTED-BRANCH extensions.

- U04 anchors to L3: pretraining, SFT, LoRA, data/loss are

OFFICIAL-SYLLABUS. Quantization, hardware efficiency, numerical precision, full-vs-PEFT comparison, regression tests, and resource budgets are REQUESTED-BRANCH (hardware trade-offs are named in L5).

- U05 anchors to L3/L4: RLHF, DPO, reward design, policy gradients, PPO

are OFFICIAL-SYLLABUS. Pairwise-label protocol detail, policy ratios, KL estimators, advantages, PPO variants, DPO derivation assumptions, reference-model mechanics, optimization pathologies, and evaluation protocol are REQUESTED-BRANCH extensions.

Second builder: U06-U09 (built 2026-10-07)

- U06 anchors to L3/L4/L5: reasoning models, verifiable outcomes

(RLVR), GRPO are OFFICIAL-SYLLABUS (L4 "reasoning with GRPO (RLVR)"). Reward-hacking limitations are OFFICIAL-SYLLABUS (L4 "limitations"). Group normalization mechanics, sampling diversity, train-time vs test-time scaling, scaling-data analysis, verifier quality, compute allocation, ablations, and generalization tests are REQUESTED-BRANCH extensions.

- U07 anchors to L6: retrieval pipeline, context construction

(context compaction), function calling and tool results (tool calling), and agent state (memory) are OFFICIAL-SYLLABUS. Advanced retrieval, hybrid search, reranking, ReAct, stopping rules, permissions, and failure separation are REQUESTED-BRANCH extensions.

- U08 anchors to L7: judge rubrics, pointwise/pairwise judging,

position/verbosity bias, human checks, appropriate metrics, and benchmark interpretation are OFFICIAL-SYLLABUS. Calibration, nondeterminism, blind evaluation, contamination, uncertainty, and cost/latency are REQUESTED-BRANCH extensions.

- U09 anchors to L8/L9 plus both exams: diffusion LLM topics

(continuous, discrete, masked diffusion, training, inference) and the L9 recap are OFFICIAL-SYLLABUS. Exam objective mapping is original practice only (both exams fell after baseline, no exam content claimed). Dated update separation, original exam-style practice, cross-lecture dependencies, comparative architectures, end-to-end toy, research protocol, production bridge, and oral defense are REQUESTED-BRANCH.

- Consolidation: 2 capstones (research replication with honest

negative result, applied/FDE with hypothetical numbers), transfer sets, oral defenses, crash course, cheatsheet.

Prerequisites

Shared bridges P01-P24 live at `../shared/prerequisites/` and are linked, never rebuilt. Each unit lesson opens with a local remediation block for its own prerequisites. Read the bridge file first if the diagnostic item for it is not full marks.

Per-unit prerequisites

- U01 (NLP to transformer foundations): P03 vectors, P05 calculus, P06

probability, P11 neural nets, P13 language modelling. Bridges: `../shared/prerequisites/p03_vectors.md`, `../shared/prerequisites/p05_calculus.md`, `../shared/prerequisites/p06_probability.md`, `../shared/prerequisites/p11_neural_nets.md`, `../shared/prerequisites/p13_language.md`.

- U02 (attention variants, positions): P04 spectral/numerical linear

algebra, P12 PyTorch and stability, P14 transformer mechanics. Bridges: `../shared/prerequisites/p04_spectral.md`, `../shared/prerequisites/p12_pytorch.md`, `../shared/prerequisites/p14_transformer.md`.

- U03 (LLM architecture and generation): P13 language modelling, P14

transformer mechanics, P15 hardware. Bridges: `../shared/prerequisites/p13_language.md`, `../shared/prerequisites/p14_transformer.md`, `../shared/prerequisites/p15_hardware.md`.

- U04 (training and adaptation): P10 ML foundations, P11 neural nets,

P12 PyTorch and stability, P15 hardware. Bridges: `../shared/prerequisites/p10_ml_foundations.md`, `../shared/prerequisites/p11_neural_nets.md`, `../shared/prerequisites/p12_pytorch.md`, `../shared/prerequisites/p15_hardware.md`.

- U05 (preference tuning, policy optimization): P08 information theory,

P17 reinforcement learning, P18 Bayesian inference and sampling. Bridges: `../shared/prerequisites/p08_information.md`, `../shared/prerequisites/p17_rl.md`, `../shared/prerequisites/p18_bayesian.md`.

Local remediation

Each lesson file opens with a remediation block: 3-5 worked micro checks that repair the exact prerequisite skills the unit needs (matrix shapes, softmax arithmetic, log rules, KL definition, policy gradient intuition). Do the block before the concepts if any diagnostic item below scores below full marks.

Diagnostic (closed-book, 15 minutes)

Answer in keys/diagnostic_key.md (separate file). Score each 0/1/2.

- D1. Compute the dot product of $[1, 2, 3]$ and $[4, -1, 0]$, and the norm of each vector.
- D2. Softmax of $[2, 1, 0]$ by hand to 3 decimals.
- D3. d/dx of $\log(1 + \exp(x))$ at $x = 0$.
- D4. A fair coin lands heads. State $P(\text{heads})$ as a probability, then as surprisal in bits.
- D5. Write the chain rule for $P(A, B, C)$ as a product of conditionals.
- D6. One linear layer: x in $\mathbb{R}^4$, W in $\mathbb{R}^{4 \times 3}$, b in $\mathbb{R}^3$. State the shape of $Wx + b$ and count its parameters.
- D7. Cross-entropy between true $[1, 0]$ and predicted $[0.7, 0.3]$.
- D8. $KL(P||Q)$ for $P = [0.5, 0.5]$, $Q = [0.9, 0.1]$. Which direction did you compute, and why does direction matter?
- D9. A policy picks action A with probability 0.6 and gets reward 10, action B with probability 0.4 and gets reward 0. State the expected return.
- D10. 7B parameters in fp16. How many bytes is the weight file, and why?

Remediation routing: D1-D2 miss -> P03/P06 bridges, D3 -> P05, D4, D7, D8 -> P08, D5 -> P06, D6 -> P11, D9 -> P17, D10 -> P15.

Notation and shapes

One meaning per symbol across U01-U05. Shapes use B (batch), T (sequence length), V (vocabulary), d (model width), h (heads), $d_k = d / h$ (head width), L (layers), E (experts).

Core symbols

Symbol	Meaning	Shape / unit
x	input token ids	(B, T), integers in [0, V)
E	token embedding table	(V, d)
X	embedded sequence	(B, T, d)
W_q, W_k, W_v	query/key/value projections	(d, d) each (MHA)
Q, K, V	projected queries, keys, values	(B, h, T, d_k)
S	attention scores	(B, h, T, T)
A	attention weights (softmax rows)	(B, h, T, T), rows sum to 1
M	mask, additive (-inf where forbidden)	(T, T) broadcast
z	logits	(B, T, V)
p	next-token distribution	(B, T, V), rows sum to 1
T (temp)	sampling temperature	scalar > 0
theta	RoPE base angle per pair	scalar, radians
r	LoRA rank	integer, $r \ll d$
alpha	LoRA scale	scalar, effective scale α / r
s	quantization scale	scalar > 0
pi_ref	frozen reference policy (DPO/PPO)	same shape as pi_theta
pi_old	rollout policy (PPO)	same shape as pi_theta
rho	policy ratio $\pi_{\theta} / \pi_{\text{old}}$	scalar per token
A_hat	advantage estimate	scalar per token
eps	PPO clip range	scalar, often 0.2
beta	DPO/KL temperature	scalar > 0
$r(x, y)$	reward model score	scalar

Shape rules

- Attention is per head: scores are (T, T) per head, never (d, d) .
- The embedding table is (V, d) : row v is the vector for token v .
- Logits are (B, T, V) : one distribution per position.
- KV cache per layer holds K and V for past tokens: $2 \times (B, h_{kv}, T, d_k)$.
- LoRA update: $dW = B A$ with $B (d, r)$, $A (r, d)$.

Units

- Probabilities are unitless in $[0, 1]$. Logits are unitless scores.
- Bits for surprisal $(-\log_2 p)$. Nats for cross-entropy with natural log, state the base every time.
- Memory in bytes, $1 \text{ GiB} = 2^{30} \text{ bytes}$. FLOPs are counts, FLOP/s is rate.
- Angles in radians.

Glossary

One meaning per term. First use in lessons matches these definitions.

- **Token:** integer id for a text piece, the model input alphabet.
- **Vocabulary:** the full set of token ids, size V .
- **Embedding:** dense vector for a token, row of the table E .
- **Logit:** unnormalized score per vocabulary item before softmax.
- **Perplexity:** exp of mean negative log-likelihood per token, lower is better, only comparable under one tokenizer.
- **Teacher forcing:** training with ground-truth previous tokens as inputs.
- **Causal mask:** mask that forbids attending to future positions.
- **KV cache:** stored keys and values of past tokens reused at decode.
- **MHA:** multi-head attention, each query head has a private KV head.
- **MQA:** multi-query attention, all query heads share one KV head.
- **GQA:** grouped-query attention, groups of query heads share KV heads.
- **RoPE:** rotary position embedding, rotates Q/K pairs by position.
- **Temperature:** divisor on logits before softmax, controls sharpness.
- **Top-k:** keep only the k highest-probability tokens, renormalize.
- **Top-p:** keep the smallest set with cumulative mass $\geq p$, renormalize.
- **ICL:** in-context learning, task behavior from prompt examples only.
- **SFT:** supervised fine-tuning on instruction-response pairs.
- **LoRA:** low-rank adapter, frozen base plus trainable BA factors.
- **PEFT:** parameter-efficient fine-tuning, trains a small subset.
- **Reward model:** scalar head scoring response quality.

- **RLHF**: pipeline SFT -> reward model -> RL against the reward model.
- **PPO**: proximal policy optimization, clipped-surrogate RL.
- **DPO**: direct preference optimization, preference loss without RL.
- **KL penalty**: term pulling the policy toward a reference policy.
- **Advantage**: return minus baseline, centers the learning signal.
- **Reward hacking**: policy exploits flaws in the reward, not the task.
- **RLVR**: RL with verifiable rewards, program-computed not judged.
- **GRPO**: group relative policy optimization, critic-free RL on prompt groups.
- **Group normalization**: $(r - \mu)/\sigma$ over one prompt group.
- **pass@k**: $1 - (1 - p)^k$, success probability over k attempts.
- **Verifier**: program that scores outcomes, e.g. unit tests.
- **Reward hacking**: policy exploits verifier gaps, not the task.
- **RAG**: retrieval-augmented generation, index then read.
- **BM25**: keyword score from term statistics.
- **RRF**: reciprocal rank fusion, $\sum 1/(k + \text{rank})$.
- **Reranker**: cross-encoder scoring (query, doc) pairs.
- **ReAct**: Thought -> Action -> Observation agent loop.
- **Tool tier**: green/yellow/red permission badge on a tool.
- **Judge rubric**: the scoring law an LLM judge follows.
- **Position bias**: judge favors the first-presented answer.
- **Verbosity bias**: judge favors the longer answer.
- **ECE**: expected calibration error, honesty about doubt.
- **Contamination**: benchmark items leaked into training.
- **Masked diffusion**: LM that unmask tokens by confidence in

parallel steps.

- **Keystone concept:** concept with the most downstream dependents.
- **Orphan objective:** exam objective with no lesson home.

Cheatsheet

One page per unit. Computed toys, not measurements. Baseline 2026-10-06.

Formulas

What	Formula
Attention	$\text{softmax}(QK^T / \sqrt{d}) V$
BT loss	$-\log \text{sigmoid}(r_w - r_l)$
PPO objective	$E[\min(\rho A, \text{clip}(\rho, 1 - \epsilon, 1 + \epsilon) A)]$
DPO loss	$-\log \text{sigmoid}(\beta \times \text{margin})$
GRPO advantage	$A_i = (r_i - \mu) / \sigma$
pass@k	$1 - (1 - p)^k$
Recall / precision	$H/R, H/K$
RRF	$\text{sum } 1/(k + \text{rank}_i)$
Debiased win rate	$(w_1 + 1 - w_2)/2$
ECE	$\text{sum}_b (n_b/n) \times \text{acc}_b - \text{conf}_b $
Wilson	$\text{center } (p + z^2/2n)/(1 + z^2/n), \text{ half-width } z \sqrt{p(1-p)/n} + z^2/4n^2/(1 + z^2/n), z = 1.96$
LoRA params	$r(d + k) \text{ vs } dk$

Key numbers (toys)

- BT: $r_w=1.2, r_l=0.4 \rightarrow P=0.69, \text{loss}=0.371$
- PPO: $\rho=2.0, A=1.0, \epsilon=0.2 \rightarrow \text{obj}=1.2$
- DPO: margin 0.14 $\rightarrow P=0.535, \text{loss}=0.626$
- GRPO: $[1,1,0,0] \rightarrow A=[1,1,-1,-1]$
- pass@16 at $p=0.2 \rightarrow 0.972$
- Retrieval: $R=8, K=20, H=6 \rightarrow \text{recall } 0.75, \text{precision } 0.30$

- Bias: orders 0.70/0.45 -> debiased 0.625
- ECE toy -> 0.074
- $n=100, w=62 \rightarrow [0.522, 0.709]$ (Wilson)

Failure -> fix

Failure	Fix
Quadratic attention	GQA, KV cache, FlashAttention
Reward hacking	harden verifier, audit gap
KL drift	KL anchor, early stop
Position bias	judge both orders
Verbosity bias	length caps in rubric
Contamination	perplexity gap, quarantine
Runaway agent	step cap, no-repeat guard
Split-table retrieval	overlapping chunks
Tied GRPO groups	raise temperature
Saturated benchmark	retire it, build harder items

Decision rules

- DPO vs RLHF: verifier exists -> RLVR/DPO, subjective -> RLHF
- Train vs test-time: rare hard queries + verifier -> test-time
- Hybrid vs dense: part numbers + prose -> hybrid
- Pointwise vs pairwise: ranking matters -> pairwise
- Full vs PEFT: new domain or big shift -> full, else LoRA
- RAG vs long context: corpus » window -> RAG
- Ship or not: lower Wilson band clears 0.5 after debiasing

Oral ladder rungs

Define, toy, derive, implement/complexity, compare, debug, critique assumptions, design experiment/transfer.

Never forget

No naked scores. Dated claims only. Orphans are decisions. The final tests the graph, not the nodes.

Crash course

Baseline 2026-10-06. Synthesized from the nine unit lessons. Every leaf stays PLANNED / SOURCE ATTRIBUTION PENDING until artifact extraction verifies it. Numbers below are the course's computed toys, not measurements.

U01 , NLP to transformer foundations

Language becomes numbers through tokenization (BPE merges frequent pairs), embeddings (vectors per token), and training objectives (next-token prediction). RNNs read serially and forget, LSTMs gate the forgetting, attention lets every position read every other position directly. The transformer block is attention plus a feedforward network, with residual links and normalization. Causal masks enforce left-to-right order. Shapes rule everything: know (batch, seq, dim) at every step.

U02 , attention variants and positions

Multi-head attention runs h parallel attentions. MQA shares one KV head across query heads (small cache, weaker), GQA groups them (the middle path). Positions enter through RoPE: rotate q/k pairs by position-dependent angles so relative distance survives the dot product. KV caching makes decoding $O(1)$ per step after a quadratic prefill. Attention is $O(n^2 d)$: the tax on long context.

U03 , LLM architecture and generation

An LLM is a decoder-only transformer trained to predict the next token, scaled until in-context learning appears. MoE routes each token to a few experts, buying capacity at fixed compute. Sampling: temperature reshapes sharpness ($T \rightarrow 0$ is argmax, $T \rightarrow \infty$ is uniform), top-k/top-p cut the tail. Prompting, ICL, and self-consistency (vote over samples) steer behavior without training. Evaluation limits: benchmarks measure proxies, not skill.

U04 , training and efficient adaptation

Pretraining minimizes next-token loss over trillions of tokens. SFT installs the behavior format on curated pairs. LoRA freezes W and trains BA (rank r , $r(d+k)$ params vs dk). Quantization (INT8) and bf16 cut memory, the optimizer states dominate the budget. Full fine-tune is expressive and forgetful, PEFT is cheap and stable. Regression tests guard what training might break.

U05 , preference tuning and policy optimization

Human comparisons become a scalar reward via the Bradley-Terry model: $\text{loss} = -\log \text{sigmoid}(r_w - r_l)$. RLHF: SFT, reward model, PPO. PPO clips the policy ratio (eps 0.2) for a pessimistic trust-region step. Advantages (return minus baseline) cut variance. DPO skips RL: the policy itself is the reward, $\text{loss} = -\log \text{sigmoid}(\beta \times \text{margin})$. Pathologies: reward hacking, KL drift, likelihood displacement. Evaluate with win rates and human spot checks.

U06 , reasoning, RL, and scaling

RLVR trains on verifiable rewards (exact match, unit tests), no human labels. GRPO samples G chains per prompt and normalizes advantages within the group: $A = (r - \mu)/\sigma$, no critic. The verifier is the single point of failure: hack it and the policy farms the hack. Test-time scaling: $\text{pass}@k = 1 - (1-p)^k$, a verifier turns samples into answers. Split fixed budgets between training (skill) and test-time (chances) at the interior optimum. Ablate one factor at a time, probe generalization on paraphrase, domain, and verifier shifts.

U07 , RAG and agentic systems

RAG: chunk, embed, index, retrieve top- k , rerank, pack into the window. Hybrid search fuses BM25 (exact terms) with dense (meaning), normalize before fusing. Rerankers trade latency for precision@5. Agents loop Thought $\rightarrow$ Action $\rightarrow$ Observation (ReAct), with tools behind JSON schemas. State lives in the trace, session summaries, and a profile. Guards: step caps, no-repeat rules, green/yellow/red permission tiers, human approval for the red. Debug upstream first: retrieval, then tools, then the model.

U08 , LLM evaluation and judging

Write anchored rubrics or the judge invents its own law. Pairwise judging is $O(n^2)$ but stable, pointwise is $O(n)$ but needs calibration. Correct position bias by judging both orders: $(w_1 + 1 - w_2)/2$. Control verbosity with length caps. Blind the judge to model identity. Report ECE for calibration honesty, Wilson intervals for uncertainty, contamination status for every benchmark. No naked scores: every number carries its band.

U09 , trends, exams, synthesis

Masked diffusion LMs unmask by confidence in parallel steps, trading serial coherence for parallel latency. Map every exam objective to a lesson home. Orphans are study priorities. The final tests the graph: attention is the keystone (15 downstream), then policy gradients and retrieval. Production needs five gates: data match, latency, cost, shadow, rollback. Defend any concept on the 8-rung ladder: define, toy, derive, implement, compare, debug, critique, design.

The one-page spine

Tokens -> attention -> sampling -> training -> preferences -> reasoning -> retrieval -> agents -> judging. Each stage has one equation, one failure, and one fix. The equations: $\text{softmax}(QK^T / \sqrt{d})V$, $-\log \text{sigmoid}(r_w - r_l)$, $A = (r - \mu)/\sigma$, $1 - (1-p)^k$, $(w_1 + 1 - w_2)/2$. The failures: quadratic cost, reward hacking, verifier gaps, position bias, contamination. The fixes: GQA and KV cache, KL anchors and audits, swap correction, quarantine, and dated honesty about what is known.